Design and Analysis of Experiments
Problem Bank: 40 Problems

Proof-oriented exam preparation (Chs. 2–11)

Detailed solutions are provided in a separate file.

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1 Problems

Problem 1 [Ch. 2 — Pooled two-sample t-test from first principles]

Let $y_{11}, \dots, y_{1n_1} \overset{\text{iid}}{\sim} N(\mu_1, \sigma^2)$ and $y_{21}, \dots, y_{2n_2} \overset{\text{iid}}{\sim} N(\mu_2, \sigma^2)$ be independent samples.

(a) Show that $\mathrm{E}\big[\sum_j (y_{ij} - \bar{y}_i)^2\big] = (n_i - 1)\sigma^2$ without using the χ^2 distribution, and conclude that $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$ is unbiased for σ^2. Then show that S_p^2 has smaller variance than either S_1^2 or S_2^2 alone (use $\mathrm{Var}(S_i^2) = 2\sigma^4/(n_i - 1)$).

(b) Using the facts $(n_i - 1)S_i^2/\sigma^2 \sim \chi^2_{n_i-1}$ and $\bar{y}_i \perp\!\!\!\perp S_i^2$, derive the null distribution of $t_0 = \dfrac{\bar{y}_1 - \bar{y}_2}{S_p\sqrt{1/n_1 + 1/n_2}}$ and invert it to obtain a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$.

(c) If $\sigma_1^2 \neq \sigma_2^2$, derive the Satterthwaite degrees of freedom ν for $t_0' = (\bar{y}_1 - \bar{y}_2)/\sqrt{S_1^2/n_1 + S_2^2/n_2}$ by matching the first two moments of $W = S_1^2/n_1 + S_2^2/n_2$ to those of a scaled χ^2_ν.

(d) Data ($n_1 = n_2 = 6$):

$$\text{X}: 20.1,\ 19.6,\ 21.3,\ 20.8,\ 19.9,\ 20.5 \qquad \text{Y}: 21.4,\ 22.0,\ 21.1,\ 22.6,\ 21.8,\ 21.7$$

Test $H_0: \mu_X = \mu_Y$ at $\alpha = 0.05$ (pooled test), give the 95% CI, test equality of variances with the F-test, and compute the Welch df.

Problem 2 [Ch. 2 — Paired design, efficiency, and an exact permutation test]

Consider $y_{ij} = \mu_i + \beta_j + \varepsilon_{ij}$, $i = 1, 2$, $j = 1, \ldots, n$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(a) Show that $d_j = y_{1j} - y_{2j}$ does not depend on β_j, find $\mathrm{E}(d_j)$ and $\mathrm{Var}(d_j)$, and derive the paired t-test.

(b) Suppose instead that $\beta_j \overset{\text{iid}}{\sim} N(0, \sigma_\beta^2)$, independent of the errors, and that a CRD with $2n$ units (each unit carries its own β) is run instead. Compute $\mathrm{Var}(\bar{y}_1 - \bar{y}_2)$ for the CRD and $\mathrm{Var}(\bar{d})$ for the paired design, and find when pairing is more precise. Discuss the loss of error degrees of freedom.

(c) Six specimens are measured under two conditions:

$$(y_{1j}, y_{2j}): \ (12, 10),\ (15, 13),\ (11, 12),\ (18, 14),\ (14, 11),\ (13, 12).$$

Carry out the paired t-test ($\alpha = 0.05$, two-sided).

(d) Under H_0 (no treatment difference, symmetric errors) the sign of each d_j is equally likely to be $+$ or $-$. Using $T = \sum_j d_j$ as statistic, compute the *exact* one-sided and two-sided permutation (sign-flip) p-values by enumerating the 2^6 sign patterns. Compare with (c).

Problem 3 [Ch. 3 — Unbalanced one-way ANOVA: identity and expected mean squares]

Let $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, $i = 1, \dots, a$, $j = 1, \dots, n_i$, $N = \sum n_i$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, with the weighted constraint $\sum_i n_i \tau_i = 0$.

(a) Prove $\sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum_i n_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2$, showing explicitly why the cross product vanishes.

(b) Derive the least squares estimators under $\sum n_i \hat{\tau}_i = 0$ from the normal equations.

(c) Prove $\mathrm{E}(MS_E) = \sigma^2$ and $\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + \dfrac{\sum_i n_i \tau_i^2}{a - 1}$. (Hint: write $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = \tau_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$ and compute $\mathrm{E} \sum_i n_i (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2$ directly.)

(d) What changes in (c) if instead the unweighted constraint $\sum \tau_i = 0$ is used while the n_i are unequal? Express $\mathrm{E}(MS_{\text{Treatments}})$ in terms of τ_i and $\bar{\tau}_w = \sum n_i \tau_i / N$.

Problem 4 [Ch. 3 — Structure of the hat matrix in the one-way model]

For the balanced one-way model with a treatments and n replicates, order the observations by treatment and use the cell-means design matrix $\mathbf{Z} = \mathbf{I}_a \otimes \mathbf{1}_n$ (column i indicates treatment i).

(a) Show that the effects-coded $\mathbf{X}$ (with $\tau_a = -\sum_{i<a} \tau_i$) and $\mathbf{Z}$ have the same column space, and hence the same hat matrix.

(b) Show that $\mathbf{H} = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T = \frac{1}{n}\,\mathbf{I}_a \otimes \mathbf{J}_n$ (block diagonal with blocks $\frac{1}{n}\mathbf{J}_n$). Interpret $\mathbf{Hy}$.

(c) Verify directly that $\mathbf{H}$ and $\mathbf{H} - \frac{1}{N}\mathbf{J}_N$ are symmetric and idempotent, and compute $\operatorname{tr}(\mathbf{H})$, $\operatorname{tr}(\mathbf{H} - \frac{1}{N}\mathbf{J}_N)$, $\operatorname{tr}(\mathbf{I} - \mathbf{H})$.

(d) Show $(\mathbf{I} - \mathbf{H})(\mathbf{H} - \frac{1}{N}\mathbf{J}_N) = \mathbf{0}$ and explain why this gives $SS_T = SS_{\text{Treatments}} + SS_E$ with orthogonal pieces.

Problem 5 [Ch. 3 — Residuals, fitted values, and independence (Exam 1, 2025)]

In the one-way model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_N)$, $N = an$, $\boldsymbol{\beta} = (\mu, \tau_1, \ldots, \tau_{a-1})^T$, let $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$.

(a) Find $\mathrm{E}(\mathbf{e}^T\mathbf{e})$ and $\mathrm{Cov}(\hat{\mathbf{y}}, \mathbf{e})$.

(b) Writing the observations with a single index $\ell = 1, \ldots, N$, show that $\sum_{\ell=1}^{N} \mathrm{Var}(\hat{y}_\ell) = \kappa\sigma^2$ and find κ. Verify the answer by computing $\mathrm{Var}(\hat{y}_\ell)$ directly from $\hat{y}_{ij} = \bar{y}_{i\cdot}$.

(c) Show that $n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$ and $\sum_i\sum_j(y_{ij} - \bar{y}_{i\cdot})^2$ are statistically independent, (i) using the quadratic-form theorem and (ii) by an elementary argument based on the independence of $\bar{y}_i$ and S_i^2.

(d) Conclude that $F_0 = MS_{\text{Treatments}}/MS_E \sim F_{a-1,N-a,\lambda}$ and identify λ.

Problem 6 [Ch. 3 — The noncentral F and the expected value of F_0]

(a) Let $U \sim \chi^2_{k,\lambda}$, i.e. $U = \sum_{i=1}^{k} Z_i^2$ with $Z_i \overset{ind}{\sim} N(\delta_i, 1)$, $\lambda = \sum \delta_i^2$. Show that $\mathrm{E}(U) = k + \lambda$.

(b) For $V \sim \chi^2_m$, show $\mathrm{E}(1/V) = 1/(m-2)$ $(m > 2)$.

(c) Hence show that in the one-way model

$$\mathrm{E}(F_0) = \frac{N-a}{N-a-2}\Big(1 + \frac{\lambda}{a-1}\Big), \qquad \lambda = \frac{n\sum \tau_i^2}{\sigma^2}.$$

Explain why this is *not* equal to $\mathrm{E}(MS_{\mathrm{Trt}})/\mathrm{E}(MS_E)$ even under H_0, and why the heuristic "$F_0 \approx 1$ under H_0" is still sensible.

(d) For $a = 4$, $n = 5$, $\sigma = 2$ and $\tau = (-1, -1, 1, 1)$, compute λ and $\mathrm{E}(F_0)$. How large must n be for $\lambda \geq 10$?

Problem 7 [Ch. 3 — Orthogonal contrasts partition $SS_{\text{Treatments}}$]

(a) For a contrast $C = \sum_i c_i y_{i\cdot}$ ($\sum c_i = 0$, balanced, n per treatment), show that $SS_C = \dfrac{(\sum_i c_i y_{i\cdot})^2}{n \sum_i c_i^2}$ has one degree of freedom and that $SS_C/\sigma^2 \sim \chi^2_1$ under $H_0 : \sum c_i \mu_i = 0$.

(b) Two contrasts are orthogonal if $\sum_i c_i d_i = 0$. Show that C and $D = \sum d_i y_{i\cdot}$ are then independent.

(c) Show that for any complete set of $a - 1$ mutually orthogonal contrasts,

$$\sum_{k=1}^{a-1} SS_{C_k} = SS_{\text{Treatments}}.$$

(Hint: normalized contrast vectors together with $\mathbf{1}/\sqrt{a}$ form an orthonormal basis of $\mathbb{R}^a$.)

(d) The tensile data have equally spaced cotton levels with totals $y_{i\cdot} = 49, 77, 88, 108, 54$, $n = 5$, $MS_E = 8.06$ (20 df). Using orthogonal polynomial coefficients linear $(-2, -1, 0, 1, 2)$, quadratic $(2, -1, -2, -1, 2)$, cubic $(-1, 2, 0, -2, 1)$, quartic $(1, -4, 6, -4, 1)$, compute the four SS, verify they add to $SS_{\text{Treatments}} = 475.76$, and test each.

Problem 8 [Ch. 3 — Scheffé's method via the Cauchy–Schwarz inequality]

Balanced one-way model, n per treatment. For $\mathbf{c} \in \mathcal{C} = \{\mathbf{c} \neq \mathbf{0} : \sum c_i = 0\}$ let $\Gamma_{\mathbf{c}} = \sum c_i \mu_i$, $C_{\mathbf{c}} = \sum c_i \bar{y}_{i\cdot}$.

(a) Show that for every $\mathbf{c} \in \mathcal{C}$,

$$\frac{(C_{\mathbf{c}} - \Gamma_{\mathbf{c}})^2}{\frac{\sigma^2}{n}\sum c_i^2} \leq \frac{n\sum_i \left[(\bar{y}_{i\cdot} - \mu_i) - (\bar{y}_{\cdot\cdot} - \bar{\mu})\right]^2}{\sigma^2},$$

with equality for a suitable $\mathbf{c}$ (use Cauchy–Schwarz on $\sum c_i(u_i - \bar{u})$).

(b) Show that the right-hand side, after replacing σ^2 by MS_E and dividing by $a-1$, has an $F_{a-1,N-a}$ distribution (for all μ_i).

(c) Conclude that

$$P\big(|C_{\mathbf{c}} - \Gamma_{\mathbf{c}}| \leq S_\alpha(\mathbf{c}) \ \ \forall \mathbf{c} \in \mathcal{C}\big) = 1 - \alpha, \qquad S_\alpha(\mathbf{c}) = \sqrt{(a-1)F_{\alpha,a-1,N-a}}\sqrt{MS_E \sum c_i^2/n}.$$

(d) Show that the Scheffé test rejects $\Gamma_{\mathbf{c}} = 0$ for *some* contrast if and only if the overall ANOVA F-test rejects H_0 at level α.

Problem 9 [Ch. 3 — Family-wise error rate and pairwise comparisons]

(a) Prove $1-(1-\alpha)^r \le r\alpha$ and, without independence, $\alpha_{FW} \le r\alpha$. Compute α_{FW} for $r=6$ independent tests at $\alpha=0.05$.

(b) A CRD with $a=4$, $n=5$ gave

T1	18	21	20	23	18
T2	25	24	27	23	26
T3	22	20	24	21	23
T4	29	27	30	28	26

Construct the ANOVA table and test equality of means.

(c) Compare all pairs using (i) Fisher's LSD, (ii) Bonferroni ($r=6$), (iii) Tukey ($q_{0.05}(4,16)=4.05$). Present the results as underlined groupings and comment on which method is most conservative.

(d) Test the contrast $\Gamma = \mu_4 - \tfrac{1}{3}(\mu_1+\mu_2+\mu_3)$ with an ordinary t-test and with Scheffé. Which is appropriate if the contrast was suggested by looking at the data?

Problem 10 [Ch. 3 — Kruskal–Wallis and variance-stabilizing transformations]

(a) Without ties, show that the ranks satisfy $\bar{R}_{\cdot\cdot} = \frac{N+1}{2}$ and $\frac{1}{N-1}\sum_{k=1}^{N}(k-\bar{R})^2 = \frac{N(N+1)}{12}$, and hence that

$$H = \frac{1}{S^2}\Big[\sum_i \frac{R_{i\cdot}^2}{n_i} - \frac{N(N+1)^2}{4}\Big] = \frac{12}{N(N+1)}\sum_i n_i\Big(\bar{R}_{i\cdot} - \frac{N+1}{2}\Big)^2.$$

(b) Three groups: G1: 3.1, 4.5, 2.8; G2: 5.6, 6.1, 4.9; G3: 7.2, 5.9, 8.0. Compute H and test at $\alpha = 0.05$ with the χ^2 approximation.

(c) Using the delta method, show that if $\mathrm{Var}(y) = w(\mu)$ then $h(\mu) \propto \int w(\mu)^{-1/2} d\mu$ stabilizes the variance. Derive the transformation for (i) Poisson counts, (ii) $\sigma \propto \mu$, (iii) $\sigma \propto \mu^2$, (iv) general $\sigma \propto \mu^{\alpha}$.

(d) For Poisson y with large mean μ, show $\mathrm{Var}(\sqrt{y}) \approx 1/4$.

Problem 11 [Ch. 3/8 — Constraints, estimability, and regression coding (Exam 1, 2025)]

(a) In $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ the normal equations have rank a. Show that under the constraint $\tau_a = 0$ the solution is $\hat{\mu} = \bar{y}_{a\cdot}$, $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{a\cdot}$, while under $\sum \tau_i = 0$ it is $\hat{\mu} = \bar{y}_{\cdot\cdot}$, $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$.

(b) Show that $\mu+\tau_i$, $\tau_i-\tau_j$ and any contrast $\sum c_i\tau_i$ have the same estimate under both constraints, but μ and τ_i individually do not. (These are the *estimable functions*.)

(c) With effects coding for $a = 4$, $n = 2$ ($x_k = 1$ for level k, -1 for level 4, 0 otherwise, $k = 1, 2, 3$) and $\mathbf{y} = (5, 5, 0, 0, 0, 0, -5, -5)^T$, write $\mathbf{X}$, compute $\mathbf{X}^T\mathbf{X}$, $\mathbf{X}^T\mathbf{y}$, $\hat{\boldsymbol{\beta}}$, $\hat{\mathbf{y}}$ and $\mathbf{e}$.

(d) Show that the hat matrix, fitted values and SS_E are identical under effects coding and dummy coding, and explain why.

Problem 12 [Ch. 4 — RCBD expected mean squares from scratch]

For $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ (a treatments, b blocks, $\sum \tau_i = \sum \beta_j = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$):

(a) Prove the three-way decomposition $SS_T = SS_{\text{Trt}} + SS_{\text{Block}} + SS_E$ and justify the error df $(a-1)(b-1)$.

(b) Prove $\text{E}(MS_{\text{Trt}}) = \sigma^2 + \frac{b}{a-1}\sum \tau_i^2$.

(c) Prove $\text{E}(SS_E) = (a-1)(b-1)\sigma^2$ *elementwise*, by expanding $e_{ij} = (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) - (\bar{\varepsilon}_{\cdot j} - \bar{\varepsilon}_{\cdot\cdot})$ and computing every expectation, including $\text{E}(\bar{\varepsilon}_{i\cdot}\bar{\varepsilon}_{\cdot j})$.

(d) Verify (c) again using $\text{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \text{tr}(\mathbf{A}\mathbf{V}) + \boldsymbol{\mu}^T\mathbf{A}\boldsymbol{\mu}$.

(e) Derive the LSEs $\hat{\mu}, \hat{\tau}_i, \hat{\beta}_j$ and show $\text{Var}(\hat{\tau}_i - \hat{\tau}_k) = 2\sigma^2/b$.

Problem 13 [Ch. 4 — The extra-sum-of-squares F-test in the RCBD (Exam 1, 2025)]

Full model (F): $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$; reduced model (R) under $H_0 : \tau_i = 0$: $y_{ij} = \mu + \beta_j + \varepsilon_{ij}$.

(a) Find $SS_{E(F)}$ and $SS_{E(R)}$ and show algebraically that $SS_{E(R)} - SS_{E(F)} = b\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

(b) Show that $\mathbf{H}_F\mathbf{H}_R = \mathbf{H}_R\mathbf{H}_F = \mathbf{H}_R$ and that $\mathbf{H}_F - \mathbf{H}_R$ is symmetric idempotent of rank $a-1$.

(c) Compute $\boldsymbol{\xi} = (\mathbf{H}_F - \mathbf{H}_R)\mathbf{X}\boldsymbol{\beta}$ and $\mathrm{E}(SS_H)$.

(d) Prove that $F_0 = \dfrac{SS_H/(a-1)}{SS_{E(F)}/[(a-1)(b-1)]} \sim F_{a-1,(a-1)(b-1)}$ under H_0, stating which theorems are used and checking every condition.

Problem 14 [Ch. 4 — What goes wrong if blocks are ignored (Exam 1, 2025)]

Let the true model be the RCBD $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$ (treatments in $\mathbf{X}_1$, blocks in $\mathbf{X}_2$), but suppose the one-way model $\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}$ is fitted.

(a) Obtain the residual vector and show $\mathrm{E}(\mathbf{e}) = (\mathbf{I} - \mathbf{H}_1)\mathbf{X}_2\boldsymbol{\beta}_2$; evaluate it elementwise.

(b) Show that $\mathrm{E}(SS_E^{\text{one-way}}) = a(b-1)\sigma^2 + a\sum_j \beta_j^2$ and hence that MS_E is biased upward.

(c) Is $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ still unbiased? Is the F-test still valid? Explain.

(d) Let $R^2 = 1 - SS_E/SS_T$. Show $R^2_{\text{RCBD}} \geq R^2_{\text{one-way}}$ and identify the difference.

(e) For the hardness data ($SS_{\text{Trt}} = 0.385$, $SS_{\text{Block}} = 0.825$, $SS_E = 0.080$, $a = b = 4$) compute both F statistics and both R^2.

Problem 15 [Ch. 4 — Relative efficiency of blocking]

Suppose blocks are a random sample, $\beta_j \overset{\text{iid}}{\sim} N(0, \sigma^2_\beta)$, so that in a CRD with the same experimental units the error variance would be $\sigma^2_{\text{CRD}} = \sigma^2_\beta + \sigma^2$.

(a) Show that in the RCBD, $\text{E}(MS_{\text{Block}}) = \sigma^2 + a\sigma^2_\beta$ and $\text{E}(MS_E) = \sigma^2$.

(b) Show that $\hat{\sigma}^2_{\text{CRD}} = \dfrac{(b-1)MS_{\text{Block}} + b(a-1)MS_E}{ab-1}$ is (approximately) unbiased for $\sigma^2_\beta + \sigma^2$, by checking $\text{E}[(b-1)MS_{\text{Block}} + b(a-1)MS_E]$ against $(ab-1)(\sigma^2 + \sigma^2_\beta)$ up to a term of order σ^2_β.

(c) Define the relative efficiency $RE = \hat{\sigma}^2_{\text{CRD}}/MS_E$ and interpret it as a ratio of required replicates.

(d) Compute RE for the hardness data ($MS_{\text{Block}} = 0.275$, $MS_E = 0.00889$, $a = b = 4$). How many replicates per treatment would a CRD need to match the precision of the RCBD?

Problem 16 [Ch. 4 — A 4×4 Latin square from start to finish]

Rows = days, columns = operators, letters = methods (square $ABCD/BCDA/CDAB/DABC$):

	O_1	O_2	O_3	O_4
D_1	A=10	B=7	C=6	D=9
D_2	B=9	C=6	D=11	A=12
D_3	C=8	D=12	A=13	B=10
D_4	D=11	A=14	B=10	C=9

(a) State the model and prove that the cross product between the row and treatment components vanishes, using only the Latin-square property.

(b) Derive the LSE of τ_j and show $\mathrm{E}(MS_{\mathrm{Trt}}) = \sigma^2 + \frac{p}{p-1}\sum \tau_j^2$.

(c) Complete the ANOVA and test for method differences at $\alpha = 0.05$.

(d) Compute Fisher's LSD and group the methods.

(e) Why does the error have only $(p-1)(p-2)$ df? What is the danger if day$\times$method interaction exists?

Problem 17 [Ch. 4 — Combinatorics and construction of BIBDs]

(a) Prove $N = ar = bk$ and $\lambda(a-1) = r(k-1)$.

(b) Show that the "unreduced" design consisting of all $\binom{a}{k}$ subsets is a BIBD and find b, r, λ.

(c) $a = 5$ gasoline additives, $b = 5$ cars, $k = 4$ additives per car (Midterm 2, 2025): find N, r, λ and construct a design; verify λ for every pair.

(d) $a = 7$, $b = 7$, $k = 3$: find r and λ. Show that the blocks $\{1, 2, 4\}$, $\{2, 3, 5\}$, ..., obtained by adding 1 (mod 7) (cyclic development) form a BIBD.

(e) Can a BIBD with $a = 6$, $k = 4$, $b = 6$ exist? Justify with the integrality conditions.

Problem 18 [Ch. 4 — Adjusted treatment effects in the BIBD (Final 2025)]

For the BIBD model $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$ with incidence n_{ij}:

(a) Write the normal equations for μ, τ_i, β_j.

(b) Eliminate the $\hat{\beta}_j$ to show $\hat{\tau}_i = \dfrac{kQ_i}{\lambda a}$, with $Q_i = y_{i\cdot} - \frac{1}{k}\sum_j n_{ij} y_{\cdot j}$. Show $\sum Q_i = 0$.

(c) Prove $\mathrm{E}(\hat{\tau}_i) = \tau_i$ by computing $\mathrm{E}(Q_i)$, and contrast with the naive estimator $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ (compute its bias).

(d) Prove $SS_{\text{Trt(adj)}} = SS_{\text{Model}} - SS_{\text{Block}} = \dfrac{k}{\lambda a}\sum_i Q_i^2$.

Problem 19 [Ch. 4 — Variances in the BIBD and its efficiency (Midterm 2, 2025)]

(a) Show $\mathrm{Var}(Q_i) = \frac{r(k-1)}{k}\sigma^2$ by splitting kQ_i into two independent pieces.

(b) Show $\mathrm{Cov}(Q_i, Q_l) = -\frac{\lambda}{k}\sigma^2$ for $i \neq l$, and check that your answers are consistent with $\sum_i Q_i = 0$.

(c) Deduce $\mathrm{Var}(\hat{\tau}_i) = \frac{k(a-1)}{\lambda a^2}\sigma^2$ and $\mathrm{Var}(\hat{\tau}_i - \hat{\tau}_l) = \frac{2k}{\lambda a}\sigma^2$.

(d) Compare with an RCBD having the same number r of replicates per treatment, define the efficiency factor $E = \lambda a/(rk)$, and show $E < 1$ whenever $k < a$. Compute E for the catalyst design ($a = b = 4$, $k = r = 3$) and for $a = 7$, $k = 3$, $\lambda = 1$.

Problem 20 [Ch. 4 — A BIBD analysis by hand]

Four treatments in four blocks of size three ($\lambda = 2$):

Trt	B_1	B_2	B_3	B_4
1	12	14	11	–
2	15	16	–	17
3	10	–	11	13
4	–	18	15	19

(a) Compute $y_{i\cdot}$, $y_{\cdot j}$, Q_i and verify $\sum Q_i = 0$.

(b) Construct the ANOVA table with adjusted treatments and test at $\alpha = 0.05$.

(c) Give $\hat{\tau}_i$ and the standard error of $\hat{\tau}_i - \hat{\tau}_l$; compare treatments 3 and 4.

(d) Compute the *unadjusted* treatment SS and explain why fitting the model with treatments entered first versus blocks entered first (`trt+block` vs `block+trt` in `lm`) gives different sequential SS but the same SS_E.

Problem 21 [Ch. 5 — Two-factor factorial: estimation, df, EMS, independence (Midterm 2, 2025)]

Model (M1): $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk}$ with the usual constraints.

(a) Derive the LSEs from the normal equations and show $\hat{\mu}_{ij} = \bar{y}_{ij\cdot}$.

(b) Explain carefully why SS_{AB} has $(a-1)(b-1)$ degrees of freedom.

(c) Prove $\mathrm{E}(MS_{AB}) = \sigma^2 + \frac{n\sum\sum(\tau\beta)_{ij}^2}{(a-1)(b-1)}$.

(d) Show that $\hat{\mu}$ and $\hat{\beta}_j$ are independent, and compute $\mathrm{Cov}(\hat{\tau}_i, \hat{\tau}_{i'})$ for $i \neq i'$ and $\mathrm{Var}(\hat{\tau}_i)$.

(e) Show that $\widehat{(\tau\beta)}_{ij}$ is uncorrelated with $\hat{\tau}_{i'}$ for all i, i', j.

Problem 22 [Ch. 5 — Comparing models with and without interaction / replication (Midterm 2, 2025)]

Consider (M1) with interaction, (M2) additive with $n > 1$, and (M3) additive with $n = 1$.

(a) Write the error SS and df of each model, and show $SS_E^{(2)} = SS_E^{(1)} + SS_{AB}$.

(b) Assume the within-cell variation is very small. Compare the F statistics for $H_0 : \tau_i = 0$ under the three models and show $F^{(2)} \approx F^{(3)} \frac{abn-a-b+1}{(a-1)(b-1)}$.

(c) What is the expected value of $MS_E^{(2)}$ when interaction is present? What are the consequences for the test of main effects?

(d) A 2×3 factorial with $n = 2$ gave (R output): A: SS 75.00 (1 df); B: SS 32.17 (2 df); A:B: SS 0.50 (2 df); Residuals: SS 9.00 (6 df). Compute MS_E for the additive model `aov(obs~A+B)` and the new F statistics.

Problem 23 [Ch. 5 — Tukey's one-degree-of-freedom test: theory (Midterm 2, 2025)]

Model: $y_{ij} = \mu + \tau_i + \beta_j + \gamma\tau_i\beta_j + \varepsilon_{ij}$, one observation per cell.

(a) Show that $\bar{y}_{\cdot\cdot}$, $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$ are unbiased for μ, τ_i, β_j even when $\gamma \neq 0$.

(b) Derive the least squares estimator of γ (treating τ_i, β_j as known, then plugging in estimates) and show

$$SS_N = \frac{\left[\sum\sum \hat{\tau}_i\hat{\beta}_j y_{ij}\right]^2}{\sum \hat{\tau}_i^2 \sum \hat{\beta}_j^2} = \frac{ab\left[\sum\sum \hat{\tau}_i\hat{\beta}_j y_{ij}\right]^2}{SS_A\, SS_B}.$$

(c) Show that the weights $c_{ij} = \hat{\tau}_i\hat{\beta}_j$ have zero row and column sums and hence $\sum\sum c_{ij}y_{ij} = \sum\sum c_{ij}e_{ij}$ (e_{ij} the additive-model residuals).

(d) Argue that $SS_N/\sigma^2 \sim \chi^2_1$ under $H_0 : \gamma = 0$, and use the Cochran-type lemma to obtain $F_0 \sim F_{1,(a-1)(b-1)-1}$.

Problem 24 [Ch. 5 — Tukey's test in practice, and a transformation]

A 3×4 experiment with one observation per cell:

	B_1	B_2	B_3	B_4
A_1	10	12	15	11
A_2	14	17	21	15
A_3	19	23	28	21

(a) Compute $\hat{\tau}_i$, $\hat{\beta}_j$, SS_A, SS_B and the residual SS.

(b) Compute SS_N and test for non-additivity at $\alpha = 0.05$.

(c) The data look multiplicative ($y \approx \mu\,\alpha_i\,\beta_j$). Explain why $\log y$ would be additive, and why the Tukey statistic is sensitive exactly to this kind of departure (relate $\gamma\tau_i\beta_j$ to a second-order expansion of $\mu\alpha_i\beta_j$).

(d) On $\log_{10} y$, the analogous computation gives $SS_N = 1.85 \times 10^{-5}$ and residual SS $= 1.54 \times 10^{-4}$. Test again and draw a conclusion.

Problem 25 [Ch. 5 — Two-factor ANOVA by hand with simple effects]

Factor A (2 levels) and factor B (3 levels), $n = 2$:

	B_1	B_2	B_3
A_1	11, 10	13, 11	13, 15
A_2	16, 14	18, 17	24, 22

(a) Compute all SS by the hand formulas, construct the ANOVA table and test all three hypotheses ($\alpha = 0.05$).

(b) Draw (describe) the interaction plot and interpret.

(c) Using Fisher's LSD, compare the levels of B separately within A_1 and within A_2.

(d) Estimate $\widehat{(\tau\beta)}_{23}$ and give its variance. (Hint: write it as a linear combination of cell means.)

Problem 26 [Ch. 6 — The 2^2 design through contrasts]

(a) Show that $A = \frac{1}{2n}[ab + a - b - (1)]$ equals the average of the simple effects of A at the two levels of B, and also equals $\bar{y}_{A^+} - \bar{y}_{A^-}$.

(b) Starting from $SS_A = bn\sum_i(\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2$, prove $SS_A = [ab + a - b - (1)]^2/(4n)$.

(c) Show $\mathrm{Var}(\text{Contrast}) = n\sigma^2\sum c_\ell^2$ and $\mathrm{E}(\text{Contrast}_A) = n(2A_2 - 2A_1)$ in the effects model, and deduce that $t_0^2 = SS_A/MS_E \sim F_{1,4(n-1)}$ under H_0.

(d) Show that the A, B, AB contrasts are mutually orthogonal and that $SS_A + SS_B + SS_{AB}$ equals the between-cells SS.

Problem 27 [Ch. 6 — General 2^k results and the regression model]

(a) Show that in the 2^k sign table, every non-identity column sums to zero and any two distinct columns are orthogonal.

(b) Show that Effect = Contrast$/(2^{k-1}n)$, $SS = \text{Contrast}^2/(2^k n)$ and $\text{Var}(\text{Effect}) = \sigma^2/(2^{k-2}n)$.

(c) For the regression model in coded variables fitted to all $2^k n$ observations, show $\mathbf{X}^T\mathbf{X} = 2^k n\,\mathbf{I}$ and that each coefficient equals half the corresponding effect. Deduce that the regression SS of each term equals the ANOVA SS.

(d) Show that a 95% CI for an effect is Effect $\pm t_{0.025,2^k(n-1)}\sqrt{MS_E/(2^{k-2}n)}$.

Problem 28 [Ch. 6 — A 2^3 design with two replicates]

Trt	(1)	a	b	ab	c	ac	bc	abc
Rep 1	22	32	35	55	44	40	60	39
Rep 2	31	43	34	47	45	37	50	41

(a) Compute all seven contrasts, effects and sums of squares.

(b) Compute SS_T, SS_E and complete the ANOVA. Which effects are significant at $\alpha = 0.05$?

(c) Write the fitted regression model using the significant terms (and needed main effects for hierarchy).

(d) Interpret the AC interaction by computing the four $A \times C$ cell means.

Problem 29 [Ch. 7 — Confounding: theory and construction]

(a) In a 2^3 design run in two blocks with ABC confounded, add block effects δ_1, δ_2 to the model. Show that the expected contrasts of all effects except ABC are free of δ's, and find $\mathrm{E}(\mathrm{Contrast}_{ABC})$.

(b) Prove that the sign of treatment combination $x \in \{0,1\}^k$ in the column $A^{\alpha_1}\cdots K^{\alpha_k}$ is $(-1)^{\sum \alpha_i(1-x_i)}$, and conclude that $L = \sum \alpha_i x_i \pmod 2$ partitions the runs according to that sign.

(c) Show that the principal block is closed under multiplication (mod 2) and that the other block is obtained by multiplying the principal block by any element not in it.

(d) Run a 2^4 in four blocks, confounding ABC and BCD. Find the generalized interaction, construct all four blocks, and state which effects are lost.

Problem 30 [Ch. 7 — Partial confounding (Final 2025 style)]

A 2^3 factorial is run in 4 replicates, each in two blocks of size 4, confounding AB in replicate 1, AC in replicate 2, BC in replicate 3 and ABC in replicate 4.

(a) Write the blocks of each replicate using defining contrasts.

(b) Give the ANOVA skeleton (sources and df) and verify the df add to 31.

(c) Explain how SS_{AB} is computed and why its relative information is 3/4. Write $\mathrm{Var}(\widehat{AB})$ in terms of σ^2 and compare with $\mathrm{Var}(\hat{A})$.

(d) Compare with complete confounding of ABC in all four replicates: what is gained and what is lost?

Problem 31 [Ch. 8 — Properties of least squares]

For $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{X}$ of full rank $p = k+1$, $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n)$:

(a) Derive the normal equations by vector differentiation, proving the two differentiation rules you use, and show the solution is a minimum.

(b) Show $\mathrm{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$, $\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$ and $SS_E = \mathbf{y}^T\mathbf{y} - \hat{\boldsymbol{\beta}}^T\mathbf{X}^T\mathbf{y}$.

(c) Show $\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \mathbf{0}$ and $\mathrm{E}(SS_E) = (n-p)\sigma^2$; conclude that $t_0 = (\hat{\beta}_j - \beta_j)/\sqrt{MS_E\, C_{jj}} \sim t_{n-p}$.

(d) Derive the $100(1-\alpha)\%$ CI for $\mathrm{E}(y \mid \mathbf{x}_0)$ and the prediction interval for a new observation at $\mathbf{x}_0$, explaining the extra "1".

Problem 32 [Ch. 8 — Simple linear regression by hand]

$$x: 1,\ 2,\ 3,\ 4,\ 5,\ 6 \qquad y: 2.1,\ 3.9,\ 6.2,\ 7.8,\ 10.1,\ 12.2$$

(a) Compute $\mathbf{X}^T\mathbf{X}$, $(\mathbf{X}^T\mathbf{X})^{-1}$, $\mathbf{X}^T\mathbf{y}$ and $\hat{\boldsymbol{\beta}}$; check against $\hat{\beta}_1 = S_{xy}/S_{xx}$.

(b) Construct the ANOVA table and test significance of regression; verify $F_0 = t_0^2$ for the slope.

(c) Give 95% CIs for β_1 and for $\mathrm{E}(y \mid x = 3.5)$ and $\mathrm{E}(y \mid x = 6)$; explain why the second is wider.

(d) Show that $\sum_{\ell} \mathrm{Var}(\hat{y}_\ell) = 2\sigma^2$ here and relate it to $\mathrm{tr}(\mathbf{H})$.

Problem 33 [Ch. 9 — One-way random effects in matrix form (Final 2025)]

$y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, $\tau_i \overset{\text{iid}}{\sim} N(0, \sigma_\tau^2)$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, independent; $\mathbf{y} = (y_{11}, \ldots, y_{1n}, \ldots, y_{an})^T$.

(a) Derive $\text{Cov}(y_{ij}, y_{i'j'})$ in all cases and write $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{y})$ with $\mathbf{I}$ and $\mathbf{J}$ (Kronecker form).

(b) Express SS_A as $\mathbf{y}^T\mathbf{A}\mathbf{y}$ and identify $\mathbf{A}$.

(c) Show that $\mathbf{A}\boldsymbol{\Sigma}/(\sigma^2 + n\sigma_\tau^2)$ is idempotent and deduce the distribution of SS_A.

(d) Obtain $\text{E}(\mathbf{y}^T\mathbf{A}\mathbf{y})$ and similarly $\text{E}(SS_E)$; derive the ANOVA estimators of σ^2, σ_τ^2.

(e) Show $SS_A \perp\!\!\!\perp SS_E$ (check $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = \mathbf{0}$ for $SS_E = \mathbf{y}^T\mathbf{B}\mathbf{y}$) and hence $F_0 = MS_A/MS_E \sim \frac{\sigma^2 + n\sigma_\tau^2}{\sigma^2} F_{a-1, N-a}$.

Problem 34 [Ch. 9 — Variance components: estimation and intervals]

Five batches are chosen at random and three assays made on each:

B_1	74	76	75
B_2	68	71	72
B_3	75	77	77
B_4	72	74	73
B_5	79	81	77

(a) Complete the ANOVA and test $H_0 : \sigma_\tau^2 = 0$.

(b) Estimate σ^2, σ_τ^2 and the intraclass correlation.

(c) Give a 95% CI for σ^2 and for $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$, deriving the latter from the pivot.

(d) Show $P(\hat\sigma_\tau^2 < 0) = P\big(F_{a-1,N-a} < 1/(1+n\theta)\big)$, $\theta = \sigma_\tau^2/\sigma^2$, and evaluate it for $\theta = 0$, 0.5, 1 in this design.

Problem 35 [Ch. 9 — Restricted vs unrestricted mixed models]

A fixed (a levels), B random (b levels), n replicates.

(a) In the restricted model with $\sum_i (\tau\beta)_{ij} = 0$ and $\text{Var}((\tau\beta)_{ij}) = \frac{a-1}{a}\sigma^2_{\tau\beta}$, show that

$$\text{Cov}((\tau\beta)_{ij}, (\tau\beta)_{i'j}) = -\sigma^2_{\tau\beta}/a.$$

(b) Derive $\text{E}(MS_A)$, $\text{E}(MS_B)$ and $\text{E}(MS_{AB})$ in the restricted model.

(c) Derive $\text{E}(MS_A)$ and $\text{E}(MS_B)$ in the unrestricted model ($(\alpha\gamma)_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2_{\alpha\gamma})$).

(d) State the test of $H_0 : \sigma^2_\beta = 0$ and the estimator of σ^2_β under each model, and verify the restricted EMS with the EMS algorithm table.

Problem 36 [Ch. 9 — The EMS algorithm and approximate F-tests (Final 2025)]

Three-factor model $y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}$, $l = 1, \ldots, n$.

(a) Factor A fixed, B and C random (restricted model). Build the EMS table with the algorithm, showing every row of the two-way table of multipliers.

(b) Which effects can be tested with an exact F-test? Give the tests.

(c) For $H_0 : \tau_i = 0$ construct a quasi-F statistic and its Satterthwaite df.

(d) Repeat (b) for the setting A, C random, B fixed and state the test for $H_0 : \beta_j = 0$.

Problem 37 [Ch. 10 — Two-stage nested design]

Two suppliers (fixed); three batches chosen at random from each (nested); two determinations per batch:

	Supplier 1			Supplier 2		
Batch	1	2	3	1	2	3
	12	15	10	18	14	19
	13	14	12	17	16	20

(a) Write the model, prove the SS decomposition and give the df.

(b) Derive $\mathrm{E}(MS_A)$ and $\mathrm{E}(MS_{B(A)})$ directly when A is fixed and B random.

(c) Complete the ANOVA, perform the appropriate tests and estimate the variance components.

(d) What would happen if the data were (incorrectly) analyzed as a crossed 2×3 factorial? Relate $SS_{B(A)}$ to SS_B and SS_{AB}.

(e) (Final 2025) Three plants are selected at random, two leaves per plant at random, n determinations per leaf. Write the model and the ANOVA with EMS and F_0's.

Problem 38 [Ch. 10 — Split-plot design]

Two blocks (replicates); whole-plot factor A (2 levels); subplot factor B (3 levels):

	A_1			A_2		
	B_1	B_2	B_3	B_1	B_2	B_3
Block 1	10	14	17	14	18	21
Block 2	13	16	19	13	19	24

(a) Explain why this is a split-plot and not a CRD or RCBD; write model II and its constraints.

(b) Prove the df identity $rab-1 = (r-1)+(a-1)+(r-1)(a-1)+(b-1)+(a-1)(b-1)+a(r-1)(b-1)$ and build the EMS table.

(c) Complete the ANOVA with the correct error terms and test A, B and AB.

(d) Comment on the precision of the whole-plot comparison here. For a paint experiment with 4 mixes (whole plot), 3 methods (subplot) and 3 days (blocks), give the df table (Final 2025).

Problem 39 [Ch. 11 — ANACOVA: derivations]

$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$, $\sum \tau_i = 0$.

(a) Derive the LSEs, showing in particular $\hat{\beta} = E_{xy}/E_{xx}$.

(b) Show $SS_E = E_{yy} - E_{xy}^2/E_{xx}$ with $a(n-1) - 1$ df.

(c) Show $\mathrm{Var}(\hat{\beta}) = \sigma^2/E_{xx}$ and $\mathrm{Var}(\hat{\tau}_i - \hat{\tau}_l) = \sigma^2\left[\frac{2}{n} + \frac{(\bar{x}_{i\cdot} - \bar{x}_{l\cdot})^2}{E_{xx}}\right]$. (Hint: $\bar{y}_{i\cdot}$ and $\hat{\beta}$ are uncorrelated.)

(d) Derive the extra-SS F tests for $H_0 : \tau_i = 0$ and $H_0 : \beta = 0$, and show the latter equals t_0^2.

(e) Explain why the covariate must not be affected by the treatments.

Problem 40 [Ch. 11 — ANACOVA for fiber strength (Final 2025)]

Strength y and thickness x for three machines:

M_1		M_2		M_3	
y	x	y	x	y	x
36	20	40	22	35	21
41	25	48	28	37	23
39	24	39	22	42	26
42	25	45	30	34	21
49	32	44	28	32	15

(a) Compute S, T, E sums of squares and cross products, and estimate μ, τ_i, β.

(b) Test $H_0 : \tau_i = 0$ and $H_0 : \beta = 0$; compare with a one-way ANOVA that ignores x.

(c) Write the regression model and design matrix for separate slopes and test for parallel slopes.

(d) Compute the adjusted treatment means and the standard error of the difference between machines 2 and 3.